

The normal-crossing example: exact law of the phase and convergence in distribution

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Abstract

Unit 220 is the last step of Astra #24's H2-lite example. For Y_i i.i.d. $N(0,1)$ the empirical phase $Z_n = n^{-1/2} \sum_{i \leq n} Y_i$ is *exactly* standard normal for every $n \geq 1$: the characteristic function of the sum of independent variables is the product (`iIndepFun.charFun_map_sum_eq_prod`), $\varphi_{N(0,1)}(t)^n = e^{-nt^2/2} = \varphi_{N(0,n)}(t)$, and characteristic functions determine finite measures (`Measure.ext_of_charFun`); scaling by $n^{-1/2}$ is Mathlib's `gaussianReal_const_mul`. Hence $Z_n \Rightarrow Z \sim N(0,1)$ trivially, and Slutsky (`TendstoInDistribution.add_of_tendstoInMeasure_const`) with the in-probability statement of unit 219 gives

$$\mathbb{E}_{\text{post}}[e^{\theta \sqrt{n} x_0 x_1}] \implies e^{Z\theta + \theta^2/2}, \quad Z \sim N(0,1)$$

(`ncFluctMGF_tendstoInDistribution`). Measurability of the posterior moment generating function in the data comes from continuity of the model integral in the phase (dominated convergence on the box). Zero `sorry/axiom`; 9097 jobs.

1 The things formalised

```
theorem continuous_nc_integral (h : Continuous ) (N) :
  Continuous fun z => x in symBox 2, x * ncKernel N z x
theorem measurable_ncRatio : Measurable fun z => ncRatio N z
theorem map_sum_gaussian (hY : i, HasLaw (Y i) (gaussianReal 0 1) P)
  (hind : iIndepFun Y P) : P.map ( i : Fin n, Y i) = gaussianReal 0 n
theorem hasLaw_ncPhaseRV (hn : 0 < n) : HasLaw (ncPhaseRV Y n) (gaussianReal 0 1) P
theorem tendstoInDistribution_ncPhaseRV :
  TendstoInDistribution (ncPhaseRV Y) atTop id (fun _ => P) (gaussianReal 0 1)
theorem aemeasurable_ncFluctMGF :
  AEMeasurable (fun _ => ncFluctMGF n (fun i => Y i) ) P
theorem ncFluctMGF_tendstoInDistribution (h0 : 0 < 0) :
  TendstoInDistribution (fun n => ncFluctMGF n (fun i : Fin n => Y i) ) atTop
  (fun z => exp (z * + ^ 2 / 2)) (fun _ => P) (gaussianReal 0 1)
```

2 Mathematics

Nothing new analytically: the exact Gaussian law of Z_n (a two-line characteristic-function argument once Mathlib's independence API is located), the domination $|\rho(x)| K_{N,z}(x) \leq A e^{|N|(|z_0|+1)}$ for $|z - z_0| < 1$ on the box (the Gaussian factor is at most 1 and $|x_0 x_1| \leq 1$), and Slutsky's lemma in the form " $X_n \Rightarrow X, Y_n \rightarrow c$ in probability $\Rightarrow X_n + Y_n \Rightarrow X + c$ ". The statement is the fluctuation

phenomenon of §4 in one concrete model: the posterior law of $\sqrt{n}x_0x_1$ is asymptotically $N(Z_n, 1)$ with $Z_n \sim N(0, 1)$ exactly.

3 Lean notes

- After `charFun_map_sum_eq_prod` the product of identical factors is a `Pi` power: `Finset.prod_const` then `Pi.pow_apply` before `charFun_gaussianReal`, then `← Complex.exp_nat_mul, push_cast, ring`.
- The `tendsto` field of `TendstoInDistribution` compares subtypes; for an eventually constant law use `tendsto_const_nhds.congr'` with `Subtype.ext` and a `change` to the underlying measure equality (`ProbabilityMeasure.coe_mk` does not fire by `rw`).
- `gaussianReal_const_mul` produces the variance `NNReal.mk (c^2) _ * v`; discharge = 1 by `NNReal.eq + push_cast`.
- The $n = 0$ sample has no data: `fun i : Fin 0 => Y i` is constant in ω (`funext fun i => i.elim0`), so measurability at $n = 0$ is `aemeasurable_const`.

4 Status

H2-lite complete: Headlines XX (deterministic phase), XX' (in probability), XX'' (in distribution). Next: fidelity review v16 of units 218–220, then freeze per Astra #24 (E2 Mellin coefficient remains an optional appendix).